

Azure Static Website Hosting Project

Overview:

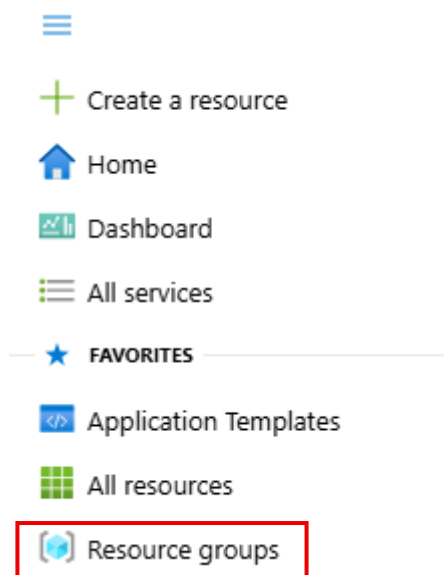
The aim of this project is to host a simple static website using Azure Blob Storage. I will initially deploy a Resource Group and storage account manually via the Azure Portal. From there, I will manually configure the Storage Account to host a static website.

Following a successful deployment, I will export the deployed infrastructure as an ARM Template and deploy a second Storage Account within a new Resource Group. A modified version of the static website will then be uploaded following the successful deployment via the ARM template.

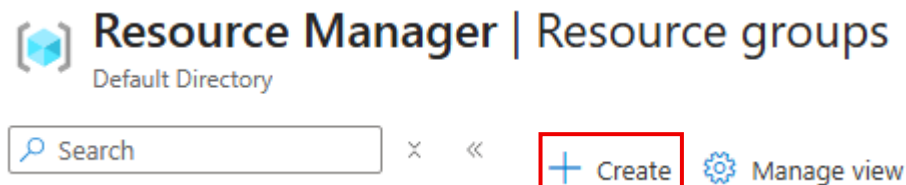
The purpose of this exercise is to reinforce my understanding of Infrastructure as Code, resource management, and Azure administration gained through my AZ-104 studies so far.

Manual Deployment:

1. Sign in to the **Azure Portal**.
2. Head to **Resource Groups**.



3. Click **Create**.



4. Select the relevant Subscription, fill in a group name and select a relevant region.

Create a resource group ...

Basics

Tags

Review + create

Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

Subscription * ⓘ

Azure subscription 1

Resource group name * ⓘ

staticweb01

Region * ⓘ

(UK) UK South

5. At the bottom, click **Review + create**.

Previous

Next

Review + create

6. Review the details of the new Resource Group.

Create a resource group ...

Basics

Tags

Review + create

[Automation Link](#)

Basics

Subscription

Azure subscription 1

Resource group name

staticweb01

Region

UK South

Tags

None

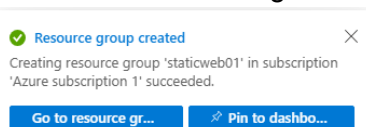
7. At the bottom again, click **Create**.

Previous

Next

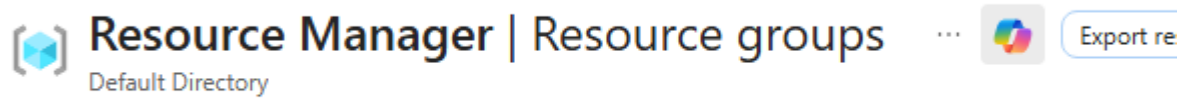
Create

A confirmation message should pop up after doing so.



8. Click **Refresh**.

Home > Resource Manager



Note that newly created Resource Group as now populated fully.

☐	Name ↑	Subscription	Location
☐	staticweb01	Azure subscription 1	UK South
☐	test01	Azure subscription 1	UK South
☐	test02	Azure subscription 1	UK South
☐	test03	Azure subscription 1	UK South

9. Head to **Home** and click **Create a resource**.

Welcome back, Tiago

Start with a template
Deploy in minutes using pre-made templates.

Create a resource
Choose a service to create a resource in your subscription.

Build an AI agent
Create and manage AI apps and agents using the latest models.

10. Select **Storage accounts**.

Popular services Free options available

From hosting and storage to functions and containers, choose the service that matches your workload and deployment needs.

Virtual machines
Build, deploy, and run your applications on resilient and scalable infrastructure.

Storage accounts
Store and access files, backups, and unstructured data reliably and securely.

SQL databases
Set up a scalable, secure relational database in minutes with built-in intelligence.

Web App
Easily host and manage websites and web applications without managing infrastructure.

Container Apps
Run your app in containers with automatic scaling and built-in microservices support.

Function App
Build serverless apps that run code on demand without worrying about servers.

Explore more

All services
Choose from over 200 services across various categories like compute, databases, analytics, and storage.

11. Select the relevant Subscription and Resource Group then fill in a unique name for the Storage Account.

Create a storage account ...

Basics Tags Review + create

 This is a simplified create experience specific to your free service selection. Don't see what you need? Go to the [standard create experience](#) for full customization of your storage account. 

Azure Storage is a Microsoft-managed service providing cloud storage that is highly available, secure, durable, scalable, and redundant. Azure Storage includes Azure Blobs (objects), Azure Data Lake Storage Gen2, Azure Files, Azure Queues, and Azure Tables. The cost of your storage account depends on the usage and the options you choose below.

[Learn more about Azure storage accounts](#) 

Project details

Select the subscription in which to create the new storage account. Choose a new or existing resource group to organize and manage your storage account together with other resources.

Subscription *	Azure subscription 1
Resource group *	staticweb01

[Create new](#)

Instance details

Storage account name * 	stwb1
Region * 	(UK) UK South

 The combination of options below is necessary for the Azure Blob Storage free service benefit. 

Performance * 	Standard: Recommended for most scenarios (general-purpose v2 account)
Redundancy * 	Locally redundant storage (LRS)

12. Click **Review + create**.

Previous	Next	Review + create
--------------------------	----------------------	---------------------------------

13. Review the details of the Storage Account.

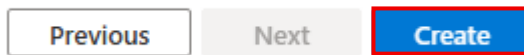
Basics Tags Review + create

[View automation template](#)

Basics

Subscription	Azure subscription 1
Resource group	staticweb01
Location	UK South
Storage account name	stwb1
Performance	Standard
Replication	Locally redundant storage (LRS)

14. Click **Create**.



15. The resource will then commence deployment.

Deployment is in progress

Deployment name : stwb1_1787501124239
Subscription : [Azure subscription 1](#)
Resource group : [staticweb01](#)

Start time : 8/23/2026, 5:08:52 PM
Correlation ID : 968ba39b-a8ab-41a8-ba9d-10cd044e0656

Deployment details

Resource	Type	Status	Operation details
 stwb1	 Storage account	Accepted	Operation details

16. Once completed, click **Go to resource**.

✓ Your deployment is complete

Deployment name : stwb1_1787501124239
Subscription : [Azure subscription 1](#)
Resource group : [staticweb01](#)

Start time : 8/23/2026, 5:08:55 PM
Correlation ID : 968ba39b-a8ab-41a8-ba9d-10cd044e0656

> Deployment details

Next steps

[Go to resource](#)

17. Head to **Data management** and then **Static website**.

Data management

 Storage Actions

 Redundancy

 Data protection

 Object replication

 Blob inventory

 Static website



 Lifecycle management

 Azure AI Search

18. Click **Enabled** and then **Save**.

[Save](#) | [Discard](#) | [Give feedback](#)

Enabling static websites on the blob service allows you to host static content. Webpages may include static content and client-side scripts. Server-side scripting is not supported. As data is replicated asynchronously from primary to secondary regions, files at the secondary endpoint may not be immediately available or in sync with files at the primary endpoint. [Learn more](#)

Static website

[Disabled](#) [Enabled](#)

Note the Primary Endpoint after saving.

Primary endpoint ⓘ

<https://stwb1.z33.web.core.windows.net/>

19. Click **\$web**.

An Azure Storage container has been created to host your static website.

\$web

20. Click **Upload**.

+ Add Directory **Upload**

\$web

21. Upload the static webpage and click **Upload**.

Upload blob



1 file(s) selected: static_page.html

Drag and drop files here or [Browse for files](#)

☐ Overwrite if files already exist

Advanced

Upload

Give feedback

22. Confirm the upload.

Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state	
☐	static_page.html	23/08/2026, 17:32:51	Hot (Inferred)	Block blob	220 B	Available	...

23. Click on the Storage Account.

Home > **stwb1 | Static website**

\$web



Container

24. Fill out the file name plus file extension under **Index document name**, then click **Save**.

Primary endpoint

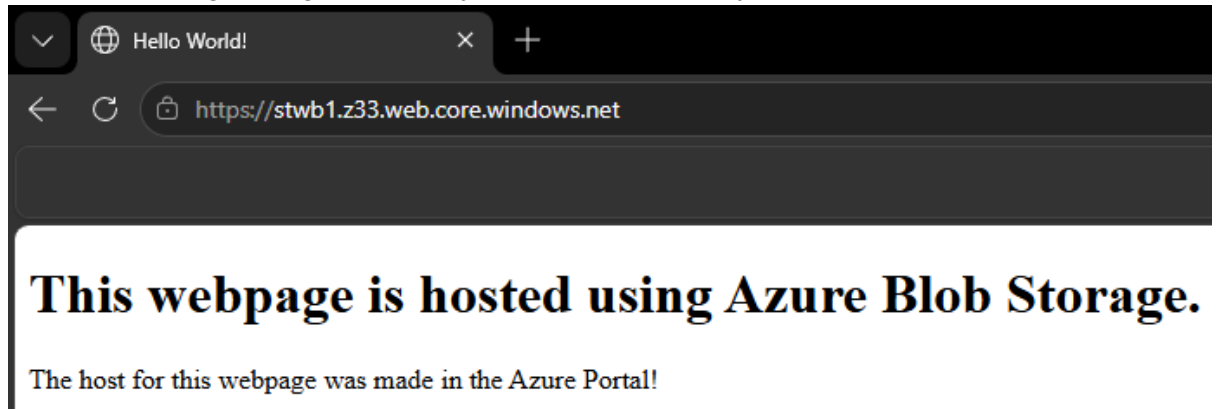
<https://stwb1.z33.web.core.windows.net/>

Index document name

static_page.html

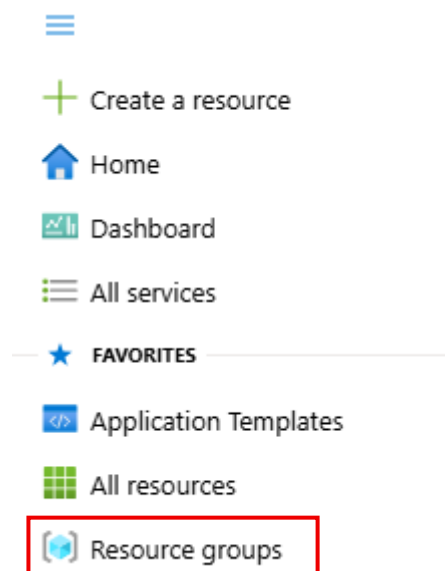


25. Test the webpage using the Primary Endpoint URL in any browser.



ARM template Deployment:

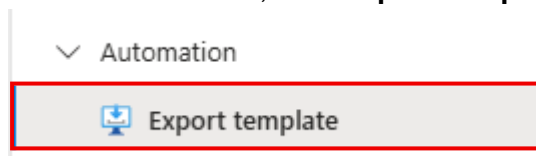
1. Sign in to the **Azure Portal**.
2. Head to **All Resources**.



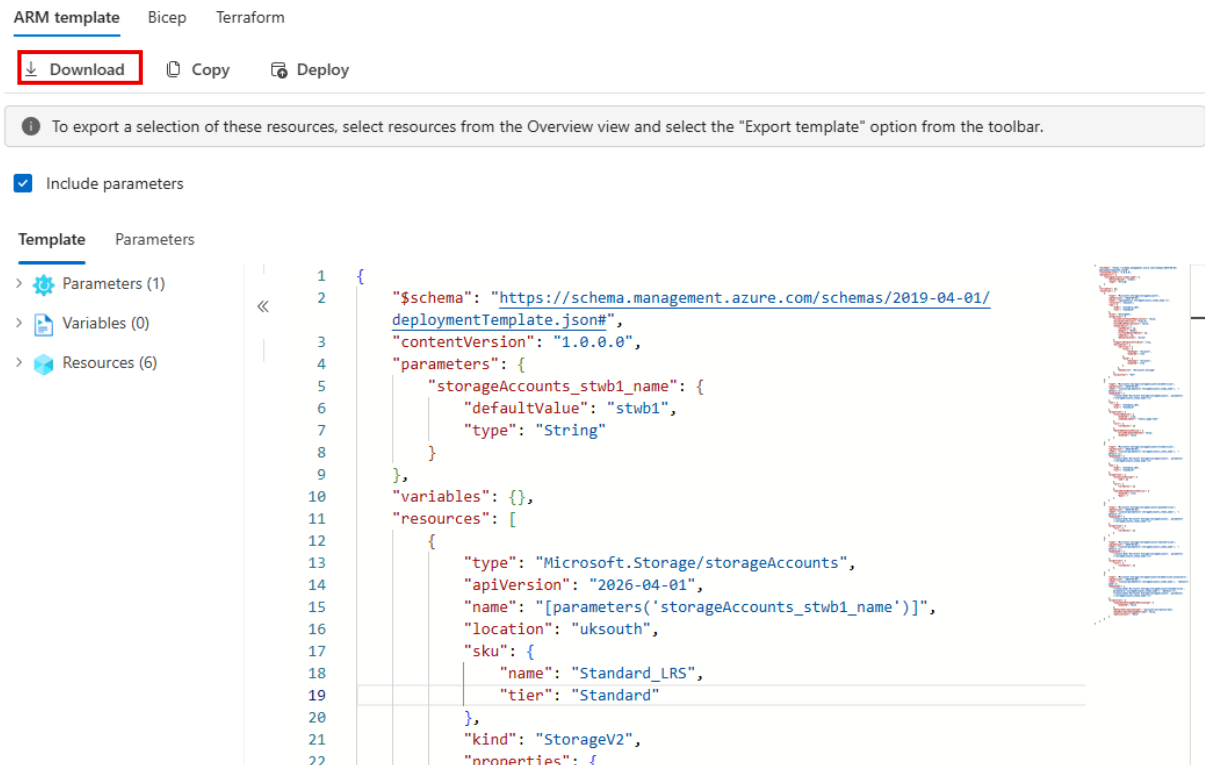
3. Click on the Resource Group of the manual deployment.

☐	Name 1	Subscription	Location
☐	staticweb01	... Azure subscription 1	UK South
☐	test01	... Azure subscription 1	UK South
☐	test02	... Azure subscription 1	UK South
☐	test03	... Azure subscription 1	UK South

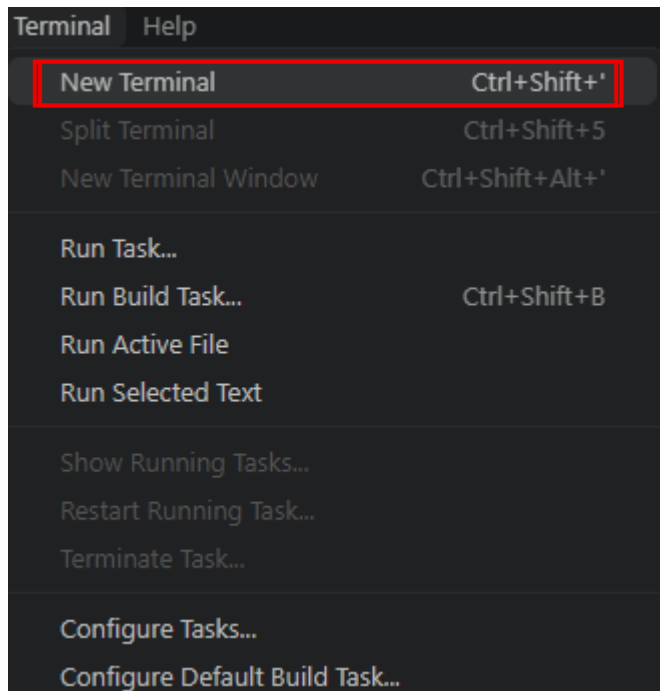
4. Under **Automation**, click **Export Template**.



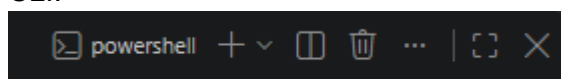
5. Under **ARM template** click Download.



6. Once downloaded, open Visual Studio code and amend the template for a second deployment.
7. Once amended, click **Terminal** and **New Terminal**.



Note: In this instance I am running PowerShell, but you can also run Git Bash for Azure CLI.



8. Run **Connect-AzAccount** and sign into Azure with an account with at least a **Contributor** role.

```
PS C:\Users\PC\OneDrive\Documents\Azure Templates> Connect-AzAccount
```

9. Now run **New-AzResourceGroup -Name <ResourceGroupName> -Location <Location>** to create a new Resource Group.

```
PS C:\Users\PC\OneDrive\Documents\Azure Templates> New-AzResourceGroup -Name staticweb02 -Location uksouth

ResourceGroupName : staticweb02
Location           : uksouth
ProvisioningState  : Succeeded
Tags               :
ResourceId         : /subscriptions/aeb8b8e6-de3d-4205-b6ab-1d24206c7645/resourceGroups/staticweb02
```

10. Run the following to begin the deployment from the ARM template:

```
$templateFile="template.json"
```

```
$today=Get-Date -Format "MM-dd-yyyy"
```

```
$deploymentName="ARMtemplate-"+$today"
```

```
New-AzResourceGroupDeployment -Name $deploymentName -TemplateFile $templateFile
```

11. Type the Resource Group name and hit **enter**.

```
cmdlet New-AzResourceGroupDeployment at command pipeline position 1
Supply values for the following parameters:
(Type !? for Help.)
ResourceGroupName: staticweb02
```

12. Wait for the prompt confirming the deployment.

```
DeploymentName      : ARMtemplate-08-23-2026
ResourceGroupName  : staticweb02
ProvisioningState   : Succeeded
Timestamp          : 23/08/2026 17:48:43
Mode               : Incremental
TemplateLink        :
Parameters          :
                    Name                               Type                               Value
                    =====                               =====
                    storageAccountName                   String                             "stwb2"

Outputs            :
DeploymentDebugLogLevel :
```

13. Head back to **Resource Groups** in the **Azure Portal** and confirm the new Resource Group.

☐	Name ↑		Subscription	Location
☐	 staticweb01	...	Azure subscription 1	UK South
☐	 staticweb02	...	Azure subscription 1	UK South
☐	 test01	...	Azure subscription 1	UK South
☐	 test02	...	Azure subscription 1	UK South
☐	 test03	...	Azure subscription 1	UK South

14. Click into the Resource Group and then click 1 Succeeded.

Deployments : 1 Succeeded
Location : UK South

15. Then click on the deployment.

☐	Deployment name	Status	Last modified	Duration	Related events
☐	ARMtemplate-08-23-2026	✓ Succeeded	8/23/2026, 6:48:43 PM	27 seconds, 875 milliseconds	Related events

16. After reviewing the deployment, **click go to resource group**.

✓ Next steps

Go to resource group

17. Click on the Storage Account.

Resources		Recommendations		
Filter for any field...		Type equals all	Location equals all	+ Add filter
☐	Name ↑	Type	Location	
☐	stwb2	Storage account	UK South	

18. Under **Data Management**, click **Static Website**.

✓ Data management

Storage Actions

Redundancy

Data protection

Object replication

Blob inventory

Static website

Lifecycle management

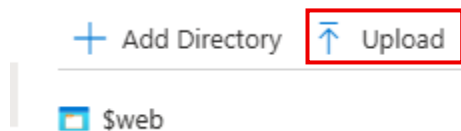
Azure AI Search

26. Click **\$web**.

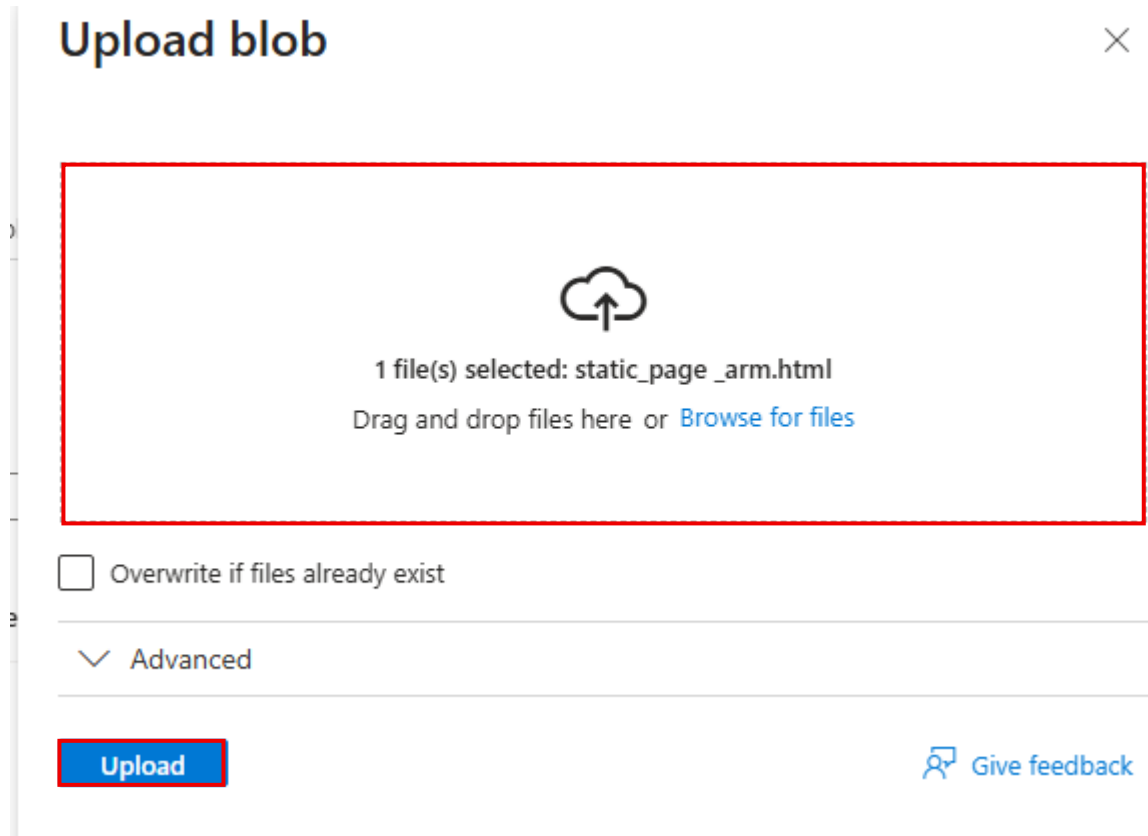
An Azure Storage container has been created to host your static website.

\$web

27. Click **Upload**.



28. Upload the static webpage and click **Upload**.



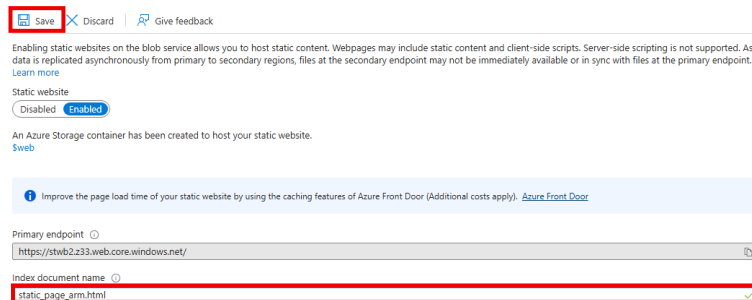
29. Confirm the upload.

☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	static_page_arm.html	23/08/2026, 20:14:17	Hot (Inferred)	Block blob	223 B	Available

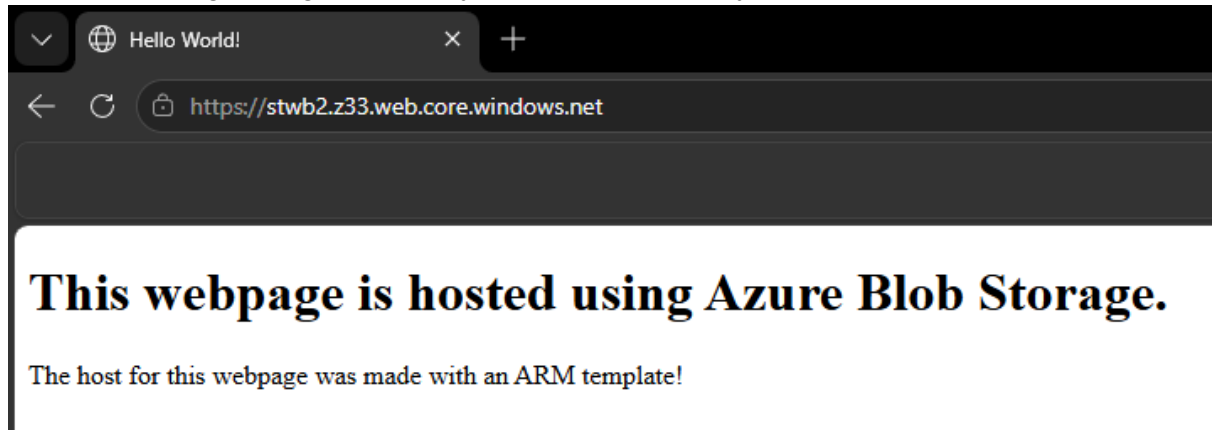
30. Click on the Storage Account.

Home > Resource Manager | All resources > stwb2 | Static website

31. Replace the **Index document name** and click **Save**.



32. Test the webpage using the Primary Endpoint URL in any browser.



Conclusion:

In summary, this project successfully demonstrated the manual deployment of Azure resources through the Azure Portal. The deployment was then exported as an ARM Template, modified, and successfully redeployed using Azure PowerShell. This exercise highlighted the importance of Azure Resource Management and Infrastructure as Code in creating consistent, repeatable, and automated deployments.